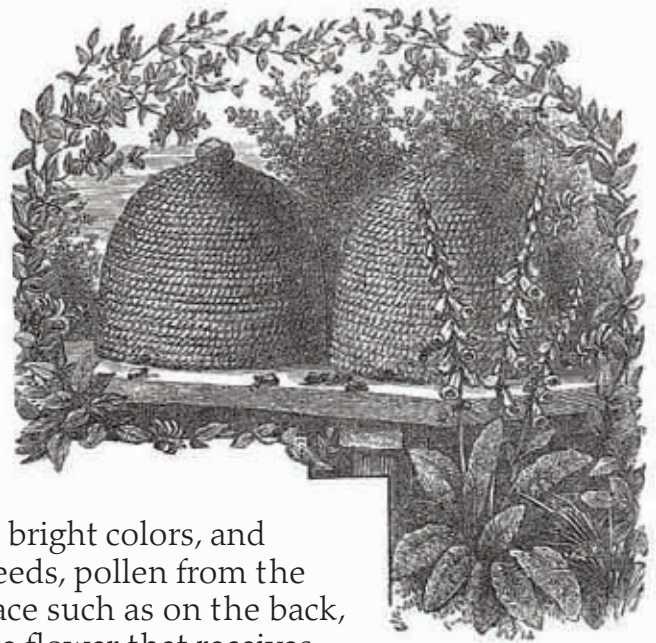


How a plant is pollinated

THE FASCINATING SHAPES and brilliant colors of many flowers have evolved over millions of years to make sure that tiny grains of pollen are carried from one plant to another. Pollen grains have to travel from the anthers to the stigma (pp. 16–17) for fertilization to occur and for seeds to be produced. Some plants are able to pollinate themselves (selfpollination), but most rely on receiving pollen from another plant of the same species (cross-pollination). Pollen may be dispersed by wind or by water, but the most important pollinators are insects. Plants entice insects to their flowers by their bright colors, and by food in the form of nectar. While the visiting insect feeds, pollen from the anthers is pressed on to its body, often at a particular place such as on the back, or on the head. The stigma of the flower that receives the pollen is in just the right place to collect it as the insect arrives. Some flowers are pollinated by a wide range of insects such as honeybees, bumblebees, hoverflies, and butterflies.



FAMILY HOME
Worker bees bring nectar and pollen back to the hive to feed the developing young.

Pollen basket on bee's hind leg

Honeyguide



THE FLORAL FEEDING STATION
Honeyguides (p. 12) on the flower guide the bee to the nectar. As the bees feed on the nectar, they also collect pollen in special baskets on their legs so that it can be carried back to the hive.

Others are more choosy and rely so heavily on a particular pollinator that no other insect species can do the job for them. Some species of yucca, for example, are pollinated exclusively by a small moth, called the yucca moth. In return, the yucca provides the moth with food and a home.

Bright yellow guide marks show bees where to land

Lower petal acts as a landing platform

OPENING UP
The flower of the common toadflax is pollinated by bumblebees. When a visiting bee arrives, the throat of the flower is tightly closed. To reach the nectar at the back of the flower, the bee must open up the flower by pushing forward.

Nectar tube

CLIMBING IN
As the bumblebee climbs over the hump that seals the flower's throat and crawls inside in search of the nectar, it brushes against the anthers inside the top of the flower. These dust its back with pollen.

FEEDING TIME
As the bee feeds on the flower's nectar, any pollen it is already carrying is transferred from its back to the stigma, and the flower is pollinated.